

SYLLABUS

COURSE NAME : FULL STACK DEVELOPMENT

COURSE CODE : MCA413-1

COURSE TYPE : THEORY

SESSION GROUP : TRIMESTER 1 and 4

DEPARTMENT : COMPUTER SCIENCE

COURSE CREDIT DETAILS

Display Name	Course Credit	Hours Per Session
Theory and Practical	4	75

DESCRIPTION

The Full Stack Development course is designed to equip students with the skills and knowledge required to build modern, scalable web applications. The curriculum covers HTML5, CSS3, JavaScript, React.js, Node.js, Express.js, MongoDB, and Next.js, providing hands-on experience in both front-end and back-end development. Students will learn to design interactive user interfaces, manage databases effectively, and develop robust server-side applications. By the end of the course, learners will be able to integrate these technologies to create dynamic, full stack solutions aligned with industry standards.

COURSE OBJECTIVES

This course is designed to gain proficiency in full stack development, acquire the skills to build dynamic and robust web applications using advanced technologies, and develop strong web programming practices. The course also enables learners to apply their knowledge to solve real-world challenges in the field of web development.

OFFERED PROGRAMMES

Master of Computer Applications (TRIMESTER -I)

COURSE OUTCOMES

Outcome Reference No.	Outcome Description	RBT Domain	Highest RBT Level
CO1	Apply HTML5 semantic features, Git for version control, and Tailwind CSS to develop	Cognitive	L3 - Apply
CO2	Implement client-side scripting using JavaScript for HTML5 APIs, forms, events, browser features, and JSON-based	Cognitive	L3 - Apply

Outcome Reference No.	Outcome Description	RBT Domain	Highest RBT Level
	data exchange.		
CO3	Develop dynamic single-page applications using React.js with hooks, routing, Axios, and state management.	Cognitive	L6 - Create
CO4	Design and optimize Node.js applications integrated with MongoDB to perform efficient CRUD operations and manage data effectively.	Cognitive	L6 - Create
CO5	Build high-performance web applications using Next.js with SSR and SSG for improved performance and SEO.	Cognitive	L6 - Create

SUSTAINABILITY DEVELOPMENT GOALS (SDG)

- Industry, innovation and infrastructure
- Quality education
- Sustainable cities

21ST CENTURY SKILLS

- Creative thinking
- Technological literacy

COURSE CONTENT

Unit No.	Unit Title	Unit Teaching Hours	Development Needs	Skill Focused	Cross Cutting Issues	Reference Link
U1	HTML5, CSS AND GITHUB	15	Global	Skill Development	Sustainability	-

Topic No.	Content	Description	Topic Reference Material Upload
U1-T1	HTML5 New Elements – HTML5 Semantics –Canvas – HTML Media –Git-commit –rollback – remote repository – GitHub – merge conflict – CSS specificity rule – Pseudo selectors – media queries – flexbox	-	-

Topic No.	Content	Description	Topic Reference Material Upload
	Lab Exercises: 1. Develop a responsive webpage layout using Tailwind CSS. Include a header, navigation bar, main content, and footer and Tailwind Typography plugin.		
U1-T2	responsive web design – transition –Tailwind responsive grid – Components (Navbar, tables, heroes, carousel, modal etc.) – font awesome icons. Introduction to Core JavaScript - Use JavaScript to interact with some of the new HTML5 apis Lab Exercises: 2. Demonstrate HTML5 APIs	-	-

Unit No.	Unit Title	Unit Teaching Hours	Development Needs	Skill Focused	Cross Cutting Issues	Reference Link
U2	CLIENT-SIDE SCRIPTING	15	Global	Skill Development	Sustainability	-

Topic No.	Content	Description	Topic Reference Material Upload
U2-T1	JavaScript objects - JS Forms – Events and Event handling-Async await-JS Navigator-JS Cookies Lab Exercises: 3. Demonstrate event handling.	-	-
U2-T2	Introduction to JSON-JSON vs XML-JSON Objects-fetch API Lab Exercises: 4. Create a web application using vanilla JavaScript with a database.	-	-

Unit No.	Unit Title	Unit Teaching Hours	Development Needs	Skill Focused	Cross Cutting Issues	Reference Link
U3	REACT.JS	15	Global	Skill	Sustainability	-

Unit No.	Unit Title	Unit Teaching Hours	Development Needs	Skill Focused	Cross Cutting Issues	Reference Link
				Development		

Topic No.	Content	Description	Topic Reference Material Upload
U3-T1	Package Manager (NPM) - ES6- Introduction to React.js - Create React App & React file structure -JSX and Components -passing and destructuring props Lab Exercises: 5. Create a React application that allows users to enter details through a form and display the profile dynamically after submission.	-	-
U3-T2	Axios - Images and Forms-Conditional Rendering - Routes - Redux Lab Exercises: . 6. Develop SPA (Single Page Application) with React.js and Implement CRUD Operation using React.js.	-	-

Unit No.	Unit Title	Unit Teaching Hours	Development Needs	Skill Focused	Cross Cutting Issues	Reference Link
U4	NODE.JS and MongoDB	15	Global	Skill Development	Sustainability	-

Topic No.	Content	Description	Topic Reference Material Upload
U4-T1	Introduction to Node.js. Express.js - Node mailer - Mongo DB Fundamentals - Documents and Collections, Databases, Query Language Lab Exercises: 7. Demonstrate Node.js file system module.	-	-
U4-T2	MongoDB CRUD Operations (Create, Read, Projection, Update, Delete) - Connecting Node.js with MongoDB - Building simple RESTful services using Express and MongoDB Lab Exercises:	-	-

Topic No.	Content	Description	Topic Reference Material Upload
	8. Build simple RESTful services using Express and MongoDB		

Unit No.	Unit Title	Unit Teaching Hours	Development Needs	Skill Focused	Cross Cutting Issues	Reference Link
U5	NEXT.JS	15	Global	Skill Development	Sustainability	-

Topic No.	Content	Description	Topic Reference Material Upload
U5-T1	Setting up Environment-initiating new Next.js Project- Routing in Next.js-Server and Client Components-Data Fetching Methods -Data Fetching Methods using React Query Lab Exercises: 9. Implement server-side rendering and client-side rendering in Next.js. Display data fetched from an API using both methods.	-	-
U5-T2	CSS in Next.js - Integrating Tailwind CSS-Mobile-First Design for responsive and mobile-first design Lab Exercises: 10. Use Tailwind CSS with Next.js to create a mobile-first, responsive webpage layout.	-	-

REFERENCE MATERIALS

Text & Reference Books

[1] HTML 5 Black Book (Covers CSS3, JavaScript, XML, XHTML, AJAX, PHP, jQuery), DT Editorial Services, Dreamtech Press 6th Edition, 2023.

[2] Modern Full-Stack Development: Using TypeScript, React, Node.js, Webpack, and Docker, Frank Zammetti, APRES, 2nd Edition, 2022.

[3] The Complete Developer: Master the Full Stack with TypeScript, React, Next.js, MongoDB, and Docker, Martin

Krause , No Starch Press, 2024.

[4] Get Programming with Node.js, Jonathan Wexler, Manning Publications, 2019.

Essential Reading

[1] Chris Northwood, The Full Stack Developer: Your Essential Guide to the Everyday Skills Expected of a Modern Full Stack Web Developer, Apress Publications, 5th Edition, 2023.

[2] Laura Lemay, Rafe Colburn & Jennifer Kyrnin, Mastering HTML, CSS & Javascript Web Publishing, BPB Publications, 4th Edition, 2022.

[3] Real-World Next.js: Build scalable, high-performance, and modern web applications using Next.js, the React framework for production, 1st Edition , Packt Publishing , 2022

Online Resources

[1] https://fullstackopen.com/en/part1/introduction_to_react

[2] <https://nodejs.org/en>

[3] <https://tailwindcss.com/docs/installation/framework-guides/nextjs>

[4] Next.js by Vercel - The React Framework

[5] <https://react.dev/>

Additional Information

EXAMINATION PATTERN

Pattern For Continuous Internal Assessment (CIA)

Examination Pattern for Continuous Internal Assessment (CIA):

C2: Descriptive Test

C4: Regular Labs Marks Consolidation

Question Paper Pattern For End Semester Examination (ESE)

Question Paper Pattern for End Term Examination (ETE):

C1: Lab Test(25 Marks)

C3: Assignment/Case Study/Others(25 Marks)

C5: 50 Marks: Descriptive/ Lab Test/Others

COURSE PLAN

COURSE FOCUS AREAS

Unit No	Unit Title	Unit Teaching Hours	Development Needs	Skills Focused	Cross Cutting Issues
U1	HTML5, CSS AND GITHUB	15	Global	Skill Development	Sustainability
U2	CLIENT-SIDE SCRIPTING	15	Global	Skill Development	Sustainability
U3	REACT.JS	15	Global	Skill Development	Sustainability
U4	NODE.JS and MongoDB	15	Global	Skill Development	Sustainability
U5	NEXT.JS	15	Global	Skill Development	Sustainability

LESSON PLAN

Week	Unit	Topics	Andragogy	Reference Resources	Additional Details
Week 1	U1	U1-T1	Case study, Experiential learning, Lecture, Project based learning		
Week 2	U1	U1-T1	Debate, Problem based learning, Product based learning		
Week 3	U1	U1-T2	Project based learning, Research based learning		AI-Assisted Learning and Assessment (AI- integrated assessment through responsible use of AI tools in software development workflows)
	U2	U2-T1	Case study, Research based learning		
Week 4	U2	U2-T1	Role plays,		

Week	Unit	Topics	Andragogy	Reference Resources	Additional Details
			Think pair and share		
	U2	U2-T2	Experiential learning, Problem based learning		
Week 5	U2	U2-T2	Concept mapping, Inquiry based learning		
Week 6	U3	U3-T1	Experiential learning, Think pair and share		
Week 7	U3	U3-T2	Experiential learning, Inquiry based learning, Lecture		
Week 8	U4	U4-T1	Experiential learning, Problem based learning, Project based learning		
Week 9	U4	U4-T2	Concept mapping, Inquiry based learning, Project based learning		
Week 10	U4	U4-T2	Case study, Experiential learning, Research based		

Week	Unit	Topics	Andragogy	Reference Resources	Additional Details
			learning		
Week 11	U5	U5-T1	Gamification, Inquiry based learning, Lecture		
Week 12	U5	U5-T2	Experiential learning, Inquiry based learning, Problem based learning		
Week 13	U5	U5-T2	Lecture		

CO-PO MAPPING

Programme 1 : Master of Computer Applications

CO-PO Mapping

CO/PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8
CO1	1	2	1	2	2	0	2	3
CO2	2	3	2	2	2	3	0	2
CO3	1	2	3	1	2	0	2	1
CO4	2	2	3	2	2	2	2	0
CO5	2	2	3	2	2	2	0	1

CO ASSESSMENT MAPPING

Assessment Rule Setting for Theory

Assessment	Assessment Levels	Total	CO1	CO2	CO3	CO4	CO5
CIA - II	Department Level	50	20	20	10	0	0
Overall	Department Level	50	10	10	10	10	10
End Term	Department Level	100	15	20	20	25	20

ASSESSMENT COMPONENT DEFINITION

Assessment Component Definition Type: Assessment Components are defined Campus Specific.

BANGALORE CENTRAL CAMPUS

Course Type: Theory

Assessment: CIA - II - Total Marks: 50

Total Marks	Scale Down To	Min Marks
50	-	-

CO Allocation		
CO1	CO2	CO3
20	20	10

Component	Description/Rubrics
Written Test	DESCRIPTION : The Mid Semester Examination (MSE) is a centralized offline theory written test conducted during Week 7 of the semester. The examination covers Unit I, Unit II, and half of Unit III and is designed to assess students' conceptual understanding and analytical abilities in front-end development and modern web application technologies. The test consists of five descriptive questions, with the last two questions offering internal choices. Each question carries 10 marks, making a total of 50 marks, and students are allotted 2 hours to complete the examination. The assessment evaluates students' ability to interpret front-end development scenarios, justify the use of HTML5, CSS/Tailwind, and Git workflows, analyze client-side scripting challenges using JavaScript, asynchronous operations, forms, and data exchange mechanisms, and examine React.js application requirements to recommend appropriate component-based design and state management strategies for Single Page Application (SPA) development. This assessment primarily measures LO1, LO2, and LO3 through application-oriented and problem-solving questions. RUBRICS : Answer key/marking scheme based on: • Accuracy of concepts • Depth of explanation • Application of knowledge • Problem-solving approach • Use of relevant examples/diagrams • Technical correctness

Assessment: Overall - Total Marks: 50

Total Marks	Scale Down To	Min Marks
50	-	-

CO Allocation				
CO1	CO2	CO3	CO4	CO5
10	10	10	10	10

Component	Description/Rubrics		
Regular Lab exercises evaluations	DESCRIPTION : CIA Component-IV evaluates students through 10 laboratory exercises covering front-end, React.js, full-stack development, and modern web application development concepts. Students are required to implement, test, document, and submit their solutions through GitHub and LMS. The assessment measures students' ability to develop responsive user interfaces, implement client-side scripting, build React-based applications, create full-stack solutions using Node.js, Express.js, and MongoDB, and develop deployment-ready applications using Next.js. Continuous evaluation focuses on practical implementation, code quality, functionality, documentation, and adherence to software development best practices.		
	RUBRICS :		
	Each program carries 10 marks and evaluation rubrics for each program would be:		
	Rubrics	Evaluation Criteria	Marks
	R1	Execution/ Implementation	2 Marks
	R2	Concept Clarity	2 Marks
	R3	Formatting and Validation	2 Marks
	R4	Complexity/Self Learning	2 Marks
R5	Github Collaboration	2 Marks	

Assessment: End Term - Total Marks: 100

Total Marks	Scale Down To	Min Marks
100	-	-

CO Allocation				
CO1	CO2	CO3	CO4	CO5
15	20	20	25	20

Component	Description/Rubrics		
Practical Test	DESCRIPTION : CIA-III evaluates students’ ability to apply full-stack development concepts through practical implementation, problem-solving, and software development activities. The assessment focuses on the design and development of responsive web applications using front-end technologies, React.js, server-side programming, database integration, and modern development frameworks. Students are required to demonstrate technical competency through coding exercises, mini-projects, GitHub-based submissions, documentation, and solution presentations. The assessment emphasizes application development, debugging, integration of multiple technologies, version control practices, and professional software engineering workflows. Continuous evaluation is carried out through practical implementation and demonstration of functional solutions, enabling students to achieve the course learning outcomes related to front-end development, full-stack integration, and modern web application design. It primarily assesses LO1, LO2, LO3, LO4, and LO5 through hands-on development and project-oriented activities.		
	RUBRICS :		
	Evaluation Rubrics for Component (Practical Test)		
	Rubrics	Components	Marks
	R1	Understanding of Problem statement and Requirements	5
	R2	Application of Concepts from the Syllabus (Checklist based evaluation)	10
R3	Code Design and Structure + Code Comments (2) Innovation / UI/UX / Responsiveness+ Validations (3)	5	
R4	Functionality and Correct	5	

Component	Description/Rubrics																	
		Output Problem Solving and Logical Reasoning (2) + Viva (3)																
Lab Examination + Written Component	DESCRIPTION : The End-Term Examination (ETE) – Component V (Lab Examination) is a practical assessment conducted during Week 12 to evaluate students’ comprehensive understanding and application of full-stack web development concepts covered across Units I to V. Students are required to design and develop a domain-based web application consisting of 3–5 interconnected pages using relevant technologies such as HTML5, CSS/Tailwind CSS, JavaScript, React.js, Next.js, database connectivity, form validation, and full-stack integration techniques. The completed application must be submitted through GitHub and LMS within the stipulated duration of 2 hours. The assessment measures students’ ability to design responsive user interfaces, implement client-side functionality, develop component-based applications, integrate server-side and database operations, and build modern full-stack web solutions following industry-standard development practices. It evaluates LO1 to LO5 through hands-on application development, problem-solving, integration, and deployment-oriented tasks. RUBRICS : <table><tr><th colspan="3">Evaluation Rubrics for Component (Practical Test + Written Component)</th></tr><tr><th>Rubrics</th><th>Components</th><th>Marks</th></tr><tr><td>R1</td><td>Understanding of Requirements (Feature sketching)</td><td>5</td></tr><tr><td>R2</td><td>Application of Concepts from the Syllabus (Checklist-based evaluation)</td><td>25</td></tr><tr><td>R3</td><td>Code Design and Structure(2) + Code Comments (2)</td><td>10</td></tr></table>			Evaluation Rubrics for Component (Practical Test + Written Component)			Rubrics	Components	Marks	R1	Understanding of Requirements (Feature sketching)	5	R2	Application of Concepts from the Syllabus (Checklist-based evaluation)	25	R3	Code Design and Structure(2) + Code Comments (2)	10
Evaluation Rubrics for Component (Practical Test + Written Component)																		
Rubrics	Components	Marks																
R1	Understanding of Requirements (Feature sketching)	5																
R2	Application of Concepts from the Syllabus (Checklist-based evaluation)	25																
R3	Code Design and Structure(2) + Code Comments (2)	10																

Component	Description/Rubrics														
		Innovation in UI/UX (2) / Responsiveness (1) + Validations (3)													
	R4	Functionality and Correct Output+ Problem Solving and Logical Reasoning (5) Github (Submission and Readme) (2) + Viva (3)	10												
MCQ, Problem Solving, Prototyping, and Document	DESCRIPTION : CIA-I is designed to evaluate students’ foundational understanding and practical application of front-end web development concepts through a combination of objective assessment and problem-solving activities. The assessment comprises two components: Component IA (MCQ Test – 10 Marks) and Component IB (Problem Solving, Prototyping, and Documentation – 15 Marks), totaling 25 Marks. The MCQ test assesses conceptual knowledge related to HTML5, CSS/Tailwind CSS, responsive design principles, and Git-based version control practices through a time-bound online examination. The problem-solving component requires students to develop a prototype-based solution for a real-world problem using HTML5, CSS, and JavaScript concepts, accompanied by technical documentation and GitHub submission. Students may responsibly utilize AI tools for self-learning, debugging, and solution enhancement while maintaining transparency regarding AI usage. The assessment evaluates students’ ability to apply front-end development concepts, implement interactive client-side functionalities, and document software solutions following professional development practices, thereby addressing LO1, LO2, and LO3. RUBRICS : <table><tr><th colspan="3">Evaluation Rubrics</th></tr><tr><th>Rubrics</th><th>Component</th><th>Marks</th></tr><tr><td>R0</td><td>MCQ Answers (Equally weighted questions)</td><td>10</td></tr><tr><td>R1</td><td>Problem definition & innovation</td><td>5</td></tr></table>			Evaluation Rubrics			Rubrics	Component	Marks	R0	MCQ Answers (Equally weighted questions)	10	R1	Problem definition & innovation	5
Evaluation Rubrics															
Rubrics	Component	Marks													
R0	MCQ Answers (Equally weighted questions)	10													
R1	Problem definition & innovation	5													

Component	Description/Rubrics		
	R2	JavaScript implementation (events/forms/fetch/JSON)	5
	R3	Documentation + GitHub quality	5

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